Ignacio de la Serna

(Publish as Ignacio Serna)

Independent AI Researcher

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(Last update: 2026.01)

EDUCATION

CALIFORNIA INSTITUTE OF TECHNOLOGY **Pasadena, CA, U.S.A.**
Visiting Student (Ph.D) Researcher **2022 - 2023**

- Advisors: Pietro Perona

AUTONOMOUS UNIVERSITY OF MADRID **Madrid, Spain**
Doctor of Philosophy in Science in ***Electrical Engineering and Computer Science (Ph.D)*** **2020 – 2024**

- Advisors: Aythami Morales (primary) and Javier Ortega-Garcia (secondary)
- Dissertation: *"Understanding Bias at Multiple Levels in Computer Vision Algorithms with a focus on Face Recognition"*

NATIONAL UNIVERSITY OF DISTANCE EDUCATION **Madrid, Spain**
Master of Science in ***Artificial Intelligence*** **2018 - 2020**

AUTONOMOUS UNIVERSITY OF MADRID **Madrid, Spain**
Bachelor of Science in ***Mathematics*** **2012 – 2018**
Bachelor of Science in ***Computer Science*** **2012 – 2018**

WORK EXPERIENCES – ACADEMIA

MAX PLANCK INSTITUTE FOR HUMAN DEVELOPMENT **Berlin, Germany**
Visiting Researcher, Center for Humans and Machines **2026.02 – present**
Post-doctoral Researcher, Center for Humans and Machines **2024.02 – 2026.01**

IE UNIVERSITY **Madrid, Spain**
Adjunct Professor, School of Science and Technology **2026.01 – present**

AUTONOMOUS UNIVERSITY OF MADRID **Madrid, Spain**
Pre-doctoral Researcher, Biometrics and Data Pattern Analytics (BiDA) Lab **2019.02 – 2020.08**

WORK EXPERIENCES -- INDUSTRY

GMV AEROSPACE AND DEFENCE S.A. **Madrid, Spain**
GNSS (Global Navigation Satellite Systems) Junior Engineer **2018.02 – 2018.09**
Research Trainee **2017.10 – 2018.02**

HONORS & DISTINCTIONS

2026 Open Philanthropy / Coefficient Giving Grant for Technical AI Safety Research ($150,000)
2020 FPI (pre-doctoral contracts Research Staff Training) scholarship of the Autonomous University of Madrid

SPEECHES & SEMINARS

Short Course | Master's Program in Artificial Intelligence
Technical University of Madrid, Spain | June 2025
Topic: *"Machine Behavior and Machine Culture"*

Invited Talk | Fairness in Biometric Systems Workshop
International Conference of Pattern Recognition (ICPR) | Montreal, Canada | August 2022
Topic: *"Where are we on measuring bias?"*

Tutorial | European Union H2020 Secure and Privacy-preserving Biometrics Summer School
Autonomous University of Madrid, Spain | July 2022
Topic: *"Hands-on: NLP Tools"*

WORK UNDER REVIEW

- ***I. Serna*** (2025). "Discovering Intersectional Bias via Directional Alignment in Face Recognition Embeddings". Under review in International Conference.
- H. Yakura, E. Lopez-Lopez, L. Brinkmann, ***I. Serna***, P. Gupta, and I. Rahwan. "Empirical evidence of Large Language Model's influence on human spoken communication", *Under review in Nature*.
- ***I. Serna***, A. Morales, and J. Fierrez. "Unraveling Machine Behavior by Multi-Level Bias Analysis and Detection: Methodology and Application to Face Biometrics", *Under review in Pattern Recognition.*

PUBLICATIONS
REFEREED JOURNALS AND CONFERENCE PAPERS
(Google Scholar link)

1. H. Yakura, E. Lopez-Lopez, L. Brinkmann, ***I. Serna***, P. Gupta, I. Soraperra, T. F. Eisenmann, I. Rahwan (April 2026). "Empirical Evidence of a Bi-Directional Linguistic Interaction between Humans and Machines", *International Conference on the Evolution of Language (EVOLANG)*, Plovdiv, Bulgaria.
2. A.H. Artiles, H. Yakura, L. Brinkmann, M.C. Sola, H.A. Alhaija, ***I. Serna***, N. Rahaman, B. Schölkopf, I. Rahwan (Dec 2025). "Cultural Alien Sampler: Open-ended art generation balancing originality and coherence", *Advances in Neural Information Processing Systems (NeurIPS), Creative AI Track*, San Diego, CA, USA.
3. M. Knott, ***I. Serna***, E. Mann and P. Perona (Feb 2025). "A Rapid Test for Accuracy and Bias of Face Recognition Technology", in *IEEE/CVF Winter Conf. on Applications of Computer Vision (WACV)*, Tucson, AZ, USA.
4. A. Hernandez, L. Brinkmann, ***I. Serna***, I., N. Rahaman, H.A. Alhaija, H. Yakura, M.C. Sola, B. Schölkopf and I. Rahwan (Dec 2024). "Alien Recombination: Exploring Concept Blends Beyond Human Cognitive Availability in Visual Art", in *"Workshop on Creativity & Generative AI." Proceedings of the Neural Information Processing Systems (NeurIPS),* Montreal, Canada.
5. A. Peña, A. Morales, J. Fierrez, ***I. Serna***, J. Ortega-Garcia, I. Puente, J. Cordova and G. Cordova (Aug 2023). "Leveraging Large Language Models for Topic Classification in the Domain of Public Affairs", in *IAPR Intl. Conf. on Document Analysis and Recognition (ICDAR) Workshops*, San José, CA, USA.
6. M. Ghafourian, J. Fierrez, R. Vera-Rodriguez, A. Morales and ***I. Serna*** (June 2023). "OTB-morph: One-time Biometrics via Morphing". Machine Intelligence Research, vol.20 no.6, pp 855-871. ISSN: 2731-5398.
7. D. DeAlcala, ***I. Serna***, A. Morales, J. Fierrez, and J. Ortega-Garcia (June 2023). "Measuring bias in

AI models: A statistical approach introducing N-Sigma", in IEEE Conf. on Computers, Software, and Applications (COMPSAC), Torino, Italia.

8. ***I. Serna***, A. Morales, J. Fierrez and N. Obradovich, (April 2022). "SensitiveLoss: Improving Accuracy and Fairness of Face Representations with Discrimination-Aware Deep Learning", *Artificial Intelligence*, vol. 305.
9. A. Morales, J. Fierrez, A. Acien, R. Tolosana and ***I. Serna***, (Feb 2022). "SetMargin Loss applied to Deep Keystroke Biometrics with Circle Packing Interpretation", *Pattern Recognition*, vol. 122.
10. ***I. Serna***, D. DeAlcala, A. Morales, J. Fierrez, J. Ortega-Garcia, (Feb 2022) "IFBiD: Inference-Free Bias Detection", in *AAAI Workshop on Artificial Intelligence Safety (SafeAI)*, Vancouver, Canada.
11. M. Ghafourian, J. Fierrez, R. Vera-Rodriguez, ***I. Serna*** and A. Morales, (Jan 2022). "OTB-morph: One-Time Biometrics via Morphing applied to Face Templates", in *IEEE/CVF Winter Conf. on Applications of Computer Vision Workshops (WACVw)*, Waikoloa, HI, USA.
12. J. Hernandez-Ortega, J. Fierrez, ***I. Serna*** and A. Morales, (Dec 2021). "FaceQgen: Semi-Supervised Deep Learning for Face Image Quality Assessment", in *IEEE Intl. Conf. on Automatic Face and Gesture Recognition (FG)*, Jodhpur, India.
13. A. Peña, ***I. Serna***, A. Morales, J. Fierrez and A. Lapedriza, "Facial Expressions as a Vulnerability in Face Recognition", in *IEEE Intl. Conf. on Image Processing (ICIP)*, Anchorage, Alaska, USA, September 2021, pp. 2988-2992.
14. ***I. Serna***, A. Peña, A. Morales and J. Fierrez, (Jan 2021). "InsideBias: Measuring Bias in Deep Networks and Application to Face Gender Biometrics", in *IAPR Intl. Conf. on Pattern Recognition (ICPR)*, Milan, Italy.
15. A. Peña, ***I. Serna***, A. Morales and J. Fierrez, (Oct 2020). "FairCVtest Demo: Understanding Bias in Multimodal Learning with a Testbed in Fair Automatic Recruitment", in *ACM Intl. Conf. on Multimodal Interaction (ICMI)*, Utrecht, The Netherlands. (also presented at ECCV 2020).
16. A. Peña, ***I. Serna***, A. Morales and J. Fierrez, (June 2020). "Bias in Multimodal AI: Testbed for Fair Automatic Recruitment", in *IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops (CVPRw)*, Seattle, USA. (also presented at ICML 2020 Workshop on Human-in-the-Loop Learning).
17. ***I. Serna***, A. Morales, J. Fierrez, M. Cebrian, N. Obradovich and I. Rahwan, (Feb 2020). "Algorithmic Discrimination: Formulation and Exploration in Deep Learning-based Face Biometrics", in *AAAI Workshop on Artificial Intelligence Safety (SafeAI)*, New York, USA.

CONFERENCE ORGANIZING COMMITTEE

<u>CONFERENCE ORGANIZING COMMITTEE</u>

- Member of Organizing Committee, IEEE / IAPR International Conference on Biometrics (ICB), 2013.
- Member of Organizing Committee, IAPR Iberian Conference on Pattern Recognition and Image Analysis (IbPRIA), 2019.

REVIEWER

- Journals:
    - IEEE Transactions on Image Processing
    - IEEE Open Journal of Signal Processing
    - IEEE Transactions on Information Forensics & Security
    - IEEE Transactions on Biometrics, Behavior, and Identity Science
    - Applied Soft Computing (Elsevier)
    - Collective Intelligence (Sage and ACM)

- The Computer Journal (British Computer Society)
- PLOS ONE
- Nature Communications Psychology / Scientific Reports
- Springer Nature AI & SOCIETY

- Conferences:
  - Computer Vision and Pattern Recognition Conference (CVPR)
  - Conference on Neural Information Processing Systems (NeurIPS)
  - International Conference on Learning Representations (ICLR)
  - International Joint Conference on Biometrics (IJCB)

PROJECT PARTICIPATION

COMPETITIVE PUBLIC PROJECTS

Project Title: HumanCAIC - Enhanced Behavioral Biometrics for Human-Centric AI in Context (TED2021-131787B-I00).
Funding Entity: European Union - NextGenerationEU Funds.
Amount Funded: 314,000 €.
Duration: September 2022 - August December 2024.
Principal Investigators: Julian Fierrez and Aythami Morales.
Description: Development and study of new human-centered learning strategies.

Project Title: BBforTAI: Biometrics and Behavior for Unbiased and Trustworthy AI with Applications (PID2021-127641OB-I00)
Funding Entity: Spanish Ministry of Economic Affairs and Digital Transformation.
Amount Funded: 273.000€.
Duration: January 2022 - December 2024.
Principal Investigators: Aythami Morales and Julián Fierrez.
Description: Development of new reliable machine learning technologies with applications based on biometric and behavioral modeling.

Project Title: BIBECA: Biometric Recognition and Behavior Analysis for a Safe and Context Sensitive Human-Machine Interaction (RTI2018-101248-B-I00R)
Funding Entity: Spanish Ministry of Economic Affairs and Digital Transformation.
Amount Funded: 240.000€.
Duration: January 2019 - December 2021.
Principal Investigators: Aythami Morales and Julián Fierrez.
Description: Development of new biometric recognition technologies based on behavior.

TECHNOLOGY TRANSFER PROJECTS TO PRIVATE SECTOR

Project Title: Detecting Fraud in Facial Biometric Systems.
Funding Entity: VERIDAS.
Duration: March 2022 - November 2022.
Amount Funded: 48,200€.
Principal Investigator: Aythami Morales Moreno.
Work performed: Development of attack detection modules in facial recognition systems.

Project Title: VINCESAI-ARGOS: Development of a Politics Intelligence Platform.
Funding Entity: Vinces Consulting.
Duration: October 2021 - November 2022.
Amount Funded: 25,380€.
Principal Investigator: Aythami Morales Moreno.

Work performed: Development of a platform for information source analysis and massive text processing.

Project Title: CECA-Analytics.
Funding Entity: Cecabank S.A..
Duration: October 2016 - September 2021,
Amount Funded: confidential (>200.000€).
Principal Investigator: Javier Ortega-García.

TEACHING

IE UNIVERSITY

CSAI.2.M.B: *AI: Machine Learning Foundations*

- Spring 2026, Enrollment: 40

AUTONOMOUS UNIVERSITY OF MADRID

EE446: Signal Analysis and Information Modeling (Undergraduate)

- Spring 2022, Enrollment: 40, Teaching Assistant (co-instructor: Javier Ortega-García)

EE52: Biometric Recognition (Masters)

- Spring 2022, Enrollment: 20, Teaching Assistant (co-instructor: Javier Ortega-García)

University Program for Seniors

- November 2023, Enrollment: 50, Instructor (co-instructor: Aythami Morales)

ADVISEES

MASTER STUDENTS

2023 – 2024	Alejandro Hernandez Artiles (CS, Technische Universität München)
2022 – 2023	Sofía Bosch Pastor (EE, Autonomous University of Madrid)
2020 – 2021	Alejandro Romero del Campo (EE, Autonomous University of Madrid)

UNDERGRADUATE STUDENTS

2022 – 2023	Javier Culebras Martínez (CS, Autonomous University of Madrid)
2021 – 2022	Rosa Hornero Romera (EE, Autonomous University of Madrid)

ADDITIONAL INFORMATION

Languages: Spanish, English, Italian